

# Methylation, Homocysteine and Choline

Folate cycle (MTHFR, MTHFD1, SHMT1), methionine cycle (MTR, MTRR, MAT1A, GNMT), transsulfuration (CBS), choline and betaine pathway (BHMT, PEMT), vitamin B12 (TCN2, FUT2, CUBN, CLYBL, MMUT) and COMT

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

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# 1. Summary

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**What was seen.** 27 SNPs were read in 20 genes (26 of chip, 1 imputed) and 1 polygenic score, plus a homocysteine score of 4 loci.

**MTHFR compound heterozygous.** Sérgio F has **C677T** (rs1801133 G/A) and **A1298C** (rs1801131 T/G), almost always in trans, which leaves about 50–60% of the activity of MTHFR. It is not homozygous C677T and does not have R594Q. It is the genetic reason for folate to always be 5-MTHF, which circumvents the step of MTHFR.

**Genetically higher homocysteine.** Three loci GWAS push it up: **DPEP1 rs154657 A/A**, **MMUT rs9369898 A/A** and **GNMT rs9296404 T/T** (imputed, DR2 1.00; rises homocysteine after a methionine load). The 4 loci score is +0.20 DP above the average, the 92nd percentile in the reference population, which is equivalent to about +0.6–0.8 µmol/L: a real but small effect. In the opposite direction, **CBS** has 2 protective alleles (rs234706 A and rs234709 T) and does not have the homocystinuria variant.

**B12 without relevant genetic risk.** MTR and MTRR with common genotypes, CUBN and CLYBL without the low B12 alleles, FUT2 heterozygous secretor; **TCN2 776G** (1 copy) gives a little less active B12. Polygenic B12 score in p34.0 (typical).

**Hill and methyl intake.** **PEMT Val175Met** and **BHMT Arg239Gln** heterozygous (need for slightly larger choline); **COMT Val/Val** (haplotype LPS, high activity: more consumption of SAM); **CPS1 C/C** (genetically lower glycine).

## 2. How to read this report

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- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality  $DR2 \geq 0.8$ . Under this threshold, or outside the imputed zone, SNP is treated as SNP not found in the data.
- **Variants:** 27 SNPs: 26 of the chip, 1 imputed and 0 not found. All genotypes were reread directly in the raw DNA and in the imputation for this report.
- **Strand:** All genotypes are in the GRCh38 strand +. 0 SNPs needed conversion to complementary strand and 2 are palindromic (A/T or C/G). When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Chromosome X:** Sérgio F has only one copy of X outside the pseudoautosomal regions, so the SNPs of X appear with a single letter (hemizygous).
- **Polygenic scores (PGS):** 1 PGS Catalog scores calculated locally with 1,082,449 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ( $n = 633$ ), calculated with exactly the same variants. Percentage  $\geq 80$ : MORE LIKELY/HIGH;  $\leq 20$ : LESS LIKELY/LOW; between 20 and 80: TYPICAL.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

### 3. Polygenic scores

Percentage in the reference population. There is no homocysteine score in the catalog with many variants; the 4 loci score described in the next section was used.

#### Polygenic score

| Trait                | Score (PGS Catalog)             | Variants used                               | Percentile | Result  | Judgement |
|----------------------|---------------------------------|---------------------------------------------|------------|---------|-----------|
| Vitamin B12 in blood | PGS003158 (Am J Hum Genet 2022) | 1,082,449 1,113,819 (weighted coverage 96%) | 34.0       | TYPICAL | typical   |

## 4. Variants by theme

For each SNP: genotype (strand +), origin, effect allele and number of copies, judgment and evidence. In the "protective" SNPs, having the allele is favourable; in the "no judgment", the allele does not have a good or bad meaning.

### Folate cycle

MTHFR forms the 5-methyl-THF (5-MTHF), the methyl donor that recycles homocysteine in methionine. MTHFD1 and SHMT1 feed the cycle with carbon units.

| Gene           | SNP       | Variant            | Genotype | Source | Effect allele (copies) | Judgement            | Allele effect                                                                       | Evidence              |
|----------------|-----------|--------------------|----------|--------|------------------------|----------------------|-------------------------------------------------------------------------------------|-----------------------|
| MTHFR          | rs1801133 | C677T (Ala222Val)  | G/A      | chip   | A (1 copy/s)           | unfavourable (light) | Thermolabile enzyme: ~35% less activity per allele; +0.16 DP homocysteine by allele | Strong                |
| MTHFR          | rs1801131 | A1298C (Glu429Ala) | T/G      | chip   | G (1 copy/s)           | unfavourable (light) | Small decrease of activity; relevant especially in heterozygous compound with C677T | Moderate (in compost) |
| MTHFR          | rs2274976 | G1793A (Arg594Gln) | C/C      | chip   | T (0 copy/s)           | favourable           | Rare variant of lower activity                                                      | Weak                  |
| MTHFD1         | rs2236225 | G1958A (Arg653Gln) | G/G      | chip   | A (0 copy/s)           | favourable           | Lower 10-formil-THF synthesis; greater choline requirement                          | Moderate              |
| SHMT1          | rs1979277 | C1420T (Leu474Phe) | G/G      | chip   | A (0 copy/s)           | favourable           | Small and inconsistent effect                                                       | Weak                  |
| SLC19A1 (RFC1) | rs1051266 | G80A (His27Arg)    | T/C      | chip   | C (1 copy/s)           | without judgment     | Reduced folate carrier; inconsistent associations                                   | Weak                  |

### Methionine cycle and methyl groups

MTR (with B12) and MTRR recycle homocysteine in methionine; MAT1A forms SAM, the universal methyl donor; GNMT spends the excess of SAM. CPS1 controls glycine.

| Gene  | SNP       | Variant            | Genotype | Source              | Effect allele (copies) | Judgement            | Allele effect                                                                                | Evidence         |
|-------|-----------|--------------------|----------|---------------------|------------------------|----------------------|----------------------------------------------------------------------------------------------|------------------|
| MTR   | rs1805087 | A2756G (Asp919Gly) | A/A      | chip                | G (0 copy/s)           | neutral              | The G allele is associated with slightly lower homocysteine (small effect)                   | Weakness         |
| MTR   | rs2275565 | intronic (GWAS)    | G/G      | chip                | T (0 copy/s)           | neutral              | The T allele low to homocysteine (−0.054 DP)                                                 | Moderate         |
| MTRR  | rs1801394 | A66G (Ile22Met)    | A/A      | chip                | G (0 copy/s)           | favourable           | Lower MTRR affinity for MTR; modest effect                                                   | Weakness         |
| MTRR  | rs1532268 | C524T (Ser175Leu)  | C/C      | chip                | T (0 copy/s)           | favourable           | Small and inconsistent effect                                                                | Weak             |
| MAT1A | rs3851059 | intronic           | G/A      | chip                | A (1 copy/s)           | unfavourable (light) | Possible lower activity of MAT1A (SAM formation); scarce functional data                     | Weak             |
| GNMT  | rs9296404 | promoter           | T/T      | imputation DR2=1.00 | T (2 copy/s)           | unfavourable         | Homocysteine higher after a methionine load (more methyl group consumption)                  | Moderate         |
| CPS1  | rs1047891 | Thr1406Asn         | C/C      | chip                | A (0 copy/s)           | neutral              | Allele A rises greatly to glycine and lower to homocysteine; C/C : genetically lower glycine | Strong (glycine) |

## Transsulfuration and clearance of thiols

Homocysteine output for cysteine and glutathione (CBS, B6-dependent) and loci GWAS that modulate homocysteine in the blood (DPEP1, MMUT).

| Gene  | SNP       | Variant            | Genotype | Source | Effect allele (copies) | Judgement    | Allele effect                                           | Evidence            |
|-------|-----------|--------------------|----------|--------|------------------------|--------------|---------------------------------------------------------|---------------------|
| CBS   | rs234706  | C699T (Tyr233Tyr)  | G/A      | chip   | A (1 copy/s)           | favourable   | Higher transsulfuration and slightly lower homocysteine | Weakness            |
| CBS   | rs234709  | intronic (GWAS)    | C/T      | chip   | T (1 copy/s)           | favourable   | The T allele low to homocysteine (−0.072 DP)            | Moderate            |
| CBS   | rs5742905 | I278T (pathogenic) | A/A      | chip   | G (0 copy/s)           | favourable   | Classical homocystinuria (recessive)                    | Strong (if present) |
| DPEP1 | rs154657  | intergenic (GWAS)  | A/A      | chip   | A (2 copy/s)           | unfavourable | +0.096 DP of homocysteine by allele                     | Moderate-strong     |
| MMUT  | rs9369898 | intergenic (GWAS)  | A/A      | chip   | A (2 copy/s)           | unfavourable | +0.045 DP homocysteine by allele                        | Moderate            |

## Collin and betaine

BHMT recycles homocysteine with betaine (alternative route to folate); PEMT manufactures choline in the liver.

| Gene | SNP        | Variant            | Genotype | Source | Effect allele (copies) | Judgement            | Allele effect                                                                                        | Evidence |
|------|------------|--------------------|----------|--------|------------------------|----------------------|------------------------------------------------------------------------------------------------------|----------|
| BHMT | rs3733890  | G742A (Arg239Gln)  | G/A      | chip   | A (1 copy/s)           | unfavourable (light) | Betaine-choline pathway; effects on inconsistent homocysteine                                        | Weak     |
| PEMT | rs7946     | Val175Met          | C/T      | chip   | T (1 copy/s)           | unfavourable (light) | Lower activity of PEMT (choline synthesis in the liver); associated with fatty liver in some studies | Weakness |
| PEMT | rs12325817 | −744G>C (promoter) | C/C      | chip   | G (0 copy/s)           | favourable           | Increased susceptibility to organic dysfunction with poor choline diet                               | Weakness |

## Vitamin B12: absorption and transport

Absorption (FUT2, CUBN), transport (TCN2) and mitochondrial use (CLYBL, MMUT) of vitamin B12.

| Gene  | SNP        | Variant                       | Genotype | Source | Effect allele (copies) | Judgement            | Allele effect                                                                             | Evidence                 |
|-------|------------|-------------------------------|----------|--------|------------------------|----------------------|-------------------------------------------------------------------------------------------|--------------------------|
| TCN2  | rs1801198  | C776G (Pro259Arg)             | G/C      | chip   | G (1 copy/s)           | unfavourable (light) | Less holo-transcobalamin (active B12) with the 776G allele; direction depends on the test | Moderate                 |
| FUT2  | rs601338   | W143X (secretor/non-secretor) | G/A      | chip   | A (1 copy/s)           | without judgment     | A/A = non-secretory (higher serum B12, different microbiome); heterozygous = secretor     | Strong (secretory state) |
| CUBN  | rs1801222  | Ser253Phe                     | G/G      | chip   | A (0 copy/s)           | favourable           | More B12 deficiency anemia                                                                | Moderate                 |
| CLYBL | rs41281112 | Arg259Ter                     | C/C      | chip   | T (0 copy/s)           | favourable           | Lower B12 (loss of function of CLYBL)                                                     | Moderate                 |

## Consumption of methyl groups (COMT)

COMT spends SAM to methylate catecholamines, estrogens and polyphenols.

| Gene | SNP    | Variant              | Genotype | Source | Effect allele (copies) | Judgement        | Allele effect                                                                                                 | Evidence          |
|------|--------|----------------------|----------|--------|------------------------|------------------|---------------------------------------------------------------------------------------------------------------|-------------------|
| COMT | rs4680 | Val158Met            | G/G      | chip   | G (2 copy/s)           | without judgment | Val (G) = high COMT activity: more consumption of SAM in catecholamine methylation, estrogens and polyphenols | Strong (activity) |
| COMT | rs6269 | promoter (haplotype) | G/G      | chip   | A (0 copy/s)           | without judgment | Haplotype marker LPS/APS/HPS                                                                                  | Moderate          |

## 5. Genetic score of homocysteine (4 loci)

| Locus | SNP       | Allele that rises homocysteine | Sérgio F copies | Allele effect  |
|-------|-----------|--------------------------------|-----------------|----------------|
| MTHFR | rs1801133 | A (677T)                       | 1               | +0.158 DP      |
| DPEP1 | rs154657  | A                              | 2               | +0.096 DP      |
| MTR   | rs2275565 | G (low T)                      | 2 G (0 T)       | −0.054 DP by T |
| MMUT  | rs9369898 | A                              | 2               | +0.045 DP      |

**Result:** +0.20 DP above the mean reference population, the 92nd percentile. In absolute value, about +0.6–0.8  $\mu\text{mol/L}$ . CBS rs234709 (1 protective allele, −0.072 DP) and GNMT T/T (effect after methionine loading) do not enter the score and pull in opposite directions.

**Interpretation:** genetics explains a small part of homocysteine; the rest depends on folate, B12, B2, choline/betaine, diet protein, renal function and drugs.

## 6. Favourable

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- **MTHFR without R594Q** (rs2274976 C/C) and without homozygous C677T.
- **CBS:** 2 Alleles that lower homocysteine (rs234706 A, rs234709 T) and without I278T (homocystinuria).
- **MTRR** A66G and C524T without the effect allele; **MTHFD1** R653Q and **SHMT1** without variant.
- **B12:** CUBN Ser253Phe and CLYBL Arg259Ter absent; typical polygenic score (p34.0).
- **PEMT –744** without the risk allele (rs12325817 C/C at strand +).

## 7. Unfavourable

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- **MTHFR heterozygous compound** C677T + A1298C (~50–60% of activity).
- **Homocysteine p92 score:** DPEP1 rs154657 A/A, MMUT rs9369898 A/A; MTR rs2275565 without the allele.
- **GNMT rs9296404 T/T** (imputed): more homocysteine after a methionine load.
- **Hill:** PEMT Val175Met and BHMT Arg239Gln heterozygous (need slightly higher).
- **TCN2 776G** (1 copy): a little less active B12.
- **MAT1A** Heterozygous rs3851059 (weak).
- **CPS1 C/C:** without the allele that rises glycine (neutral, but removes a protection).

## 8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

| Mechanism                         | Genetic finding                                                                                                           | Meaning              |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------|----------------------|
| Epigenetic Changes                | MTHFR compound heterozygous and GNMT T/T: less 5-MTHF and more consumption of SAM, the methylation donor of DNA           | unfavourable (light) |
| Epigenetic Derivative             | Availability of methyl groups (MTHFR, MTR, MAT1A, COMT Val/Val) determines the maintenance of methylation throughout life | unfavourable (light) |
| DNA Damage and DNA Repair Deficit | Less 5,10-methylene-THF for thymidine synthesis (MTHFR); MTHFD1 without variant                                           | unfavourable (light) |
| Vascular Aging                    | Genetically higher homocysteine (p92 score) damage the endothelium                                                        | unfavourable (light) |
| Oxidative stress                  | CBS with 2 protective alleles: more transsulfuration for cysteine and glutathione                                         | favourable           |
| Mitochondrial dysfunction         | MMUT A/A and CLYBL without loss of function: mitochondrial use of B12                                                     | mixed                |
| Metabolic Homeostasis Decline     | PEMT Val175Met (hepatic choline, fatty liver) and heterozygous BHMT                                                       | unfavourable (light) |
| Dysbiosis                         | Heterozygous secretor FUT2: intestinal glycans profile and microbiome                                                     | neutral              |

## 9. Practical suggestions

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All suggestions are suggestion (not applied): the real stack of Sérgio F remains as it is, and any change should be discussed with the doctor.

1. **Folate always as 5-MTHF** (never the synthetic form), as it is already in the stack: circumvents the MTHFR step. *Suggestion (not applied).*
2. **Measure homocysteine with folate in red blood cells and methylmalonic acid (MMA) or holotranscobalamin**, to separate lack of functional folate from B12 (TCN2 776G). *Suggestion (not applied).*
3. **Hill and betaine by feeding** (eggs, fish, legumes, beetroots, spinach, whole grains): PEMT and BHMT heterozygous. TMG is already in the stack. *Suggestion (not applied).*
4. **Riboflavin (B2)** is the cofactor of thermolabile MTHFR; it is already in the stack; it does not need high doses. *Suggestion (not applied).*
5. **Many fiber and green leaf vegetables** (natural folate); protein distributed throughout the day, without large loads of methionine in a single meal (GNMT T/T). *Suggestion (not applied).*
6. **Glycine and serine by feeding** (collagen, legumes, soybean), because CPS1 C/C gives genetically lower glycine. *Suggestion (not applied).*
7. **Methotrexate or nitrous oxide (anesthesia)** only with 5-MTHF support and warning to the compound MTHFR doctor. *Suggestion (not applied).*
8. **Zero alcohol** (already the rule): alcohol reduces folate and methylation. *Suggestion (not applied).*

## 10. Limitations

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- Small and added effects; homocysteine depends much more on diet, vitamins, kidneys and drugs than on these genes.
- The phase (cis/trans) of the compound MTHFR is not read in the chip; the trans is almost universal, but it is not confirmed.
- The homocysteine score uses only 4 loci; there is no score with many variants in the catalog.
- GNMT rs9296404 is imputed (DR2 1.00).
- Conclusions only from genes; do not include blood analysis.

## 11. Methods and sources

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- **Genes and SNPs:** 27 SNPs chosen from the previous analysis of methylation (51 lines), keeping the functional variants and loci GWAS homocysteine, B12 and choline; SNPs without established effect were removed.
- **Genotypes:** Sérgio F chip (GRCh38, strand +), reread for this report; GNMT comes from local imputation (Beagle 5.5, public panel of genomes complete to 30x; DR2  $\geq$  0.8). No genetic data came out of the analysis computer.
- **Guidance:** Alleles of effect on strand +; in the genes of strand – (MTHFR, CBS, PEMT) the allele of the gene was converted (e.g. C677T = A in strand +). 2 palindromic SNPs were assumed in the strand +.
- **Homocysteine score:** weighted sum of published effects (van Meurs 2013) for 4 loci, compared to the reference population.
- **Comparison with previous analysis (02-10-2026):** The 26 genotypes of the chip are equal; the GNMT imputed agrees with the previous inference.
- **Context Literature:** van Meurs 2013, Paré 2009, Zeisel (coline), Miller 2002 (TCN2).

## 12. Annex: all variants analysed

| Topic                                    | Gene           | SNP        | Genotype | Source              | Effect allele | Copies | Chain                                                          | Interpretation                                  |
|------------------------------------------|----------------|------------|----------|---------------------|---------------|--------|----------------------------------------------------------------|-------------------------------------------------|
| Folate cycle                             | MTHFR          | rs1801133  | G/A      | chip                | A             | 1      | strand +                                                       | 1 copy of risk allele                           |
| Folate cycle                             | MTHFR          | rs1801131  | T/G      | chip                | G             | 1      | strand +                                                       | 1 copy of risk allele                           |
| Folate cycle                             | MTHFR          | rs2274976  | C/C      | chip                | T             | 0      | strand +                                                       | without the risk allele                         |
| Folate cycle                             | MTHFD1         | rs2236225  | G/G      | chip                | A             | 0      | strand +                                                       | without the risk allele                         |
| Folate cycle                             | SHMT1          | rs1979277  | G/G      | chip                | A             | 0      | strand +                                                       | without the risk allele                         |
| Folate cycle                             | SLC19A1 (RFC1) | rs1051266  | T/C      | chip                | C             | 1      | strand +                                                       | 1 copy of effect allele                         |
| Methionine cycle and methyl groups       | MTR            | rs1805087  | A/A      | chip                | G             | 0      | strand +                                                       | without the protective allele (common genotype) |
| Methionine cycle and methyl groups       | MTR            | rs2275565  | G/G      | chip                | T             | 0      | strand +                                                       | without the protective allele (common genotype) |
| Methionine cycle and methyl groups       | MTRR           | rs1801394  | A/A      | chip                | G             | 0      | strand +                                                       | without the risk allele                         |
| Methionine cycle and methyl groups       | MTRR           | rs1532268  | C/C      | chip                | T             | 0      | strand +                                                       | without the risk allele                         |
| Methionine cycle and methyl groups       | MAT1A          | rs3851059  | G/A      | chip                | A             | 1      | strand +                                                       | 1 copy of risk allele                           |
| Methionine cycle and methyl groups       | GNMT           | rs9296404  | T/T      | imputation DR2=1.00 | T             | 2      | strand +                                                       | 2 risk allele copies                            |
| Methionine cycle and methyl groups       | CPS1           | rs1047891  | C/C      | chip                | A             | 0      | strand +                                                       | without the protective allele (common genotype) |
| Transsulfuration and clearance of thiols | CBS            | rs234706   | G/A      | chip                | A             | 1      | strand +                                                       | 1 copy of the protective allele                 |
| Transsulfuration and clearance of thiols | CBS            | rs234709   | C/T      | chip                | T             | 1      | strand +                                                       | 1 copy of the protective allele                 |
| Transsulfuration and clearance of thiols | CBS            | rs5742905  | A/A      | chip                | G             | 0      | homozygous for the reference allele (monomorphic site on chip) | without the risk allele                         |
| Transsulfuration and clearance of thiols | DPEP1          | rs154657   | A/A      | chip                | A             | 2      | strand +                                                       | 2 risk allele copies                            |
| Transsulfuration and clearance of thiols | MMUT           | rs9369898  | A/A      | chip                | A             | 2      | strand +                                                       | 2 risk allele copies                            |
| Collin and betaine                       | BHMT           | rs3733890  | G/A      | chip                | A             | 1      | strand +                                                       | 1 copy of risk allele                           |
| Collin and betaine                       | PEMT           | rs7946     | C/T      | chip                | T             | 1      | strand +                                                       | 1 copy of risk allele                           |
| Collin and betaine                       | PEMT           | rs12325817 | C/C      | chip                | G             | 0      | strand + (palindromic; assumed +)                              | without the risk allele                         |
| Vitamin B12: absorption and transport    | TCN2           | rs1801198  | G/C      | chip                | G             | 1      | strand + (palindromic; assumed +)                              | 1 copy of risk allele                           |

| Topic                                 | Gene  | SNP        | Genotype | Source | Effect allele | Copies | Chain    | Interpretation                              |
|---------------------------------------|-------|------------|----------|--------|---------------|--------|----------|---------------------------------------------|
| Vitamin B12: absorption and transport | FUT2  | rs601338   | G/A      | chip   | A             | 1      | strand + | 1 copy of effect allele                     |
| Vitamin B12: absorption and transport | CUBN  | rs1801222  | G/G      | chip   | A             | 0      | strand + | without the risk allele                     |
| Vitamin B12: absorption and transport | CLYBL | rs41281112 | C/C      | chip   | T             | 0      | strand + | without the risk allele                     |
| Consumption of methyl groups (COMT)   | COMT  | rs4680     | G/G      | chip   | G             | 2      | strand + | Val/Val: high activity COMT (haplotype LPS) |
| Consumption of methyl groups (COMT)   | COMT  | rs6269     | G/G      | chip   | A             | 0      | strand + | 0 copy of the effect allele                 |